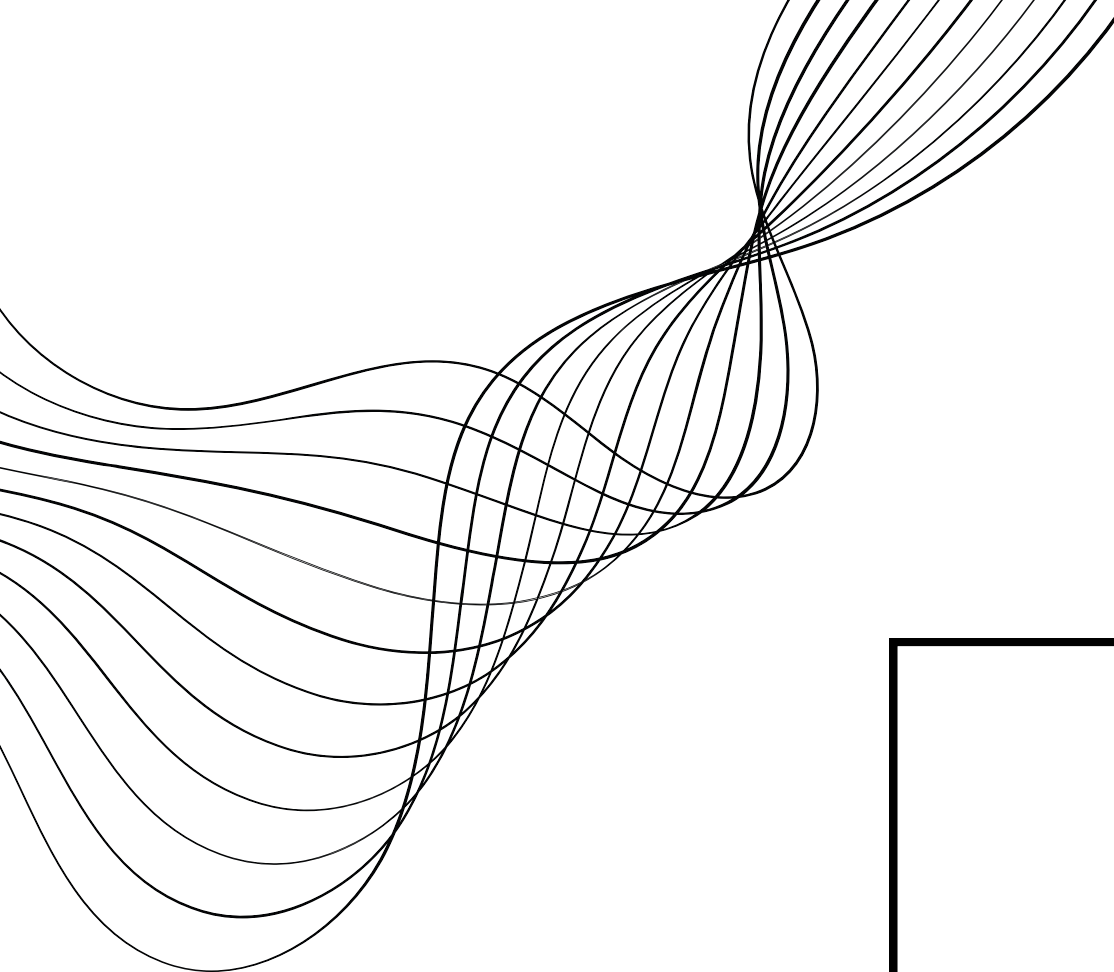
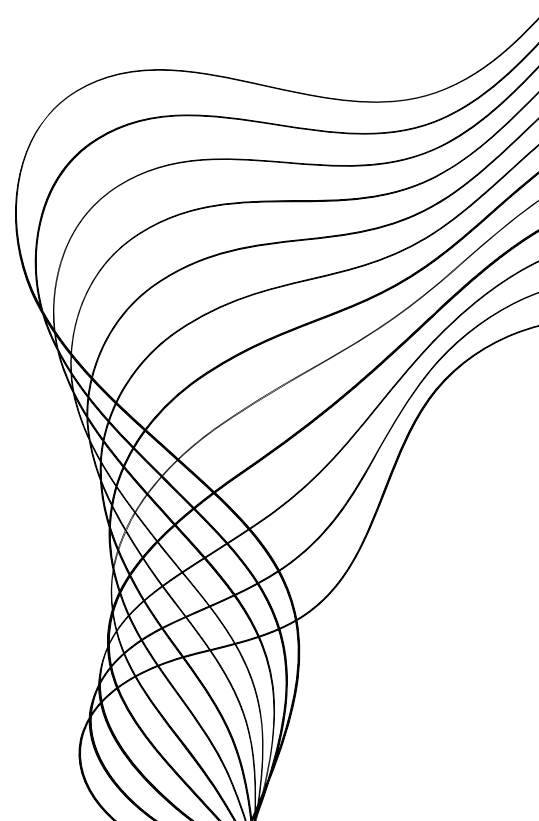




ESTIMATING NBA FREE AGENT CONTRACTS

Joshua Li, Jeffrey Ma

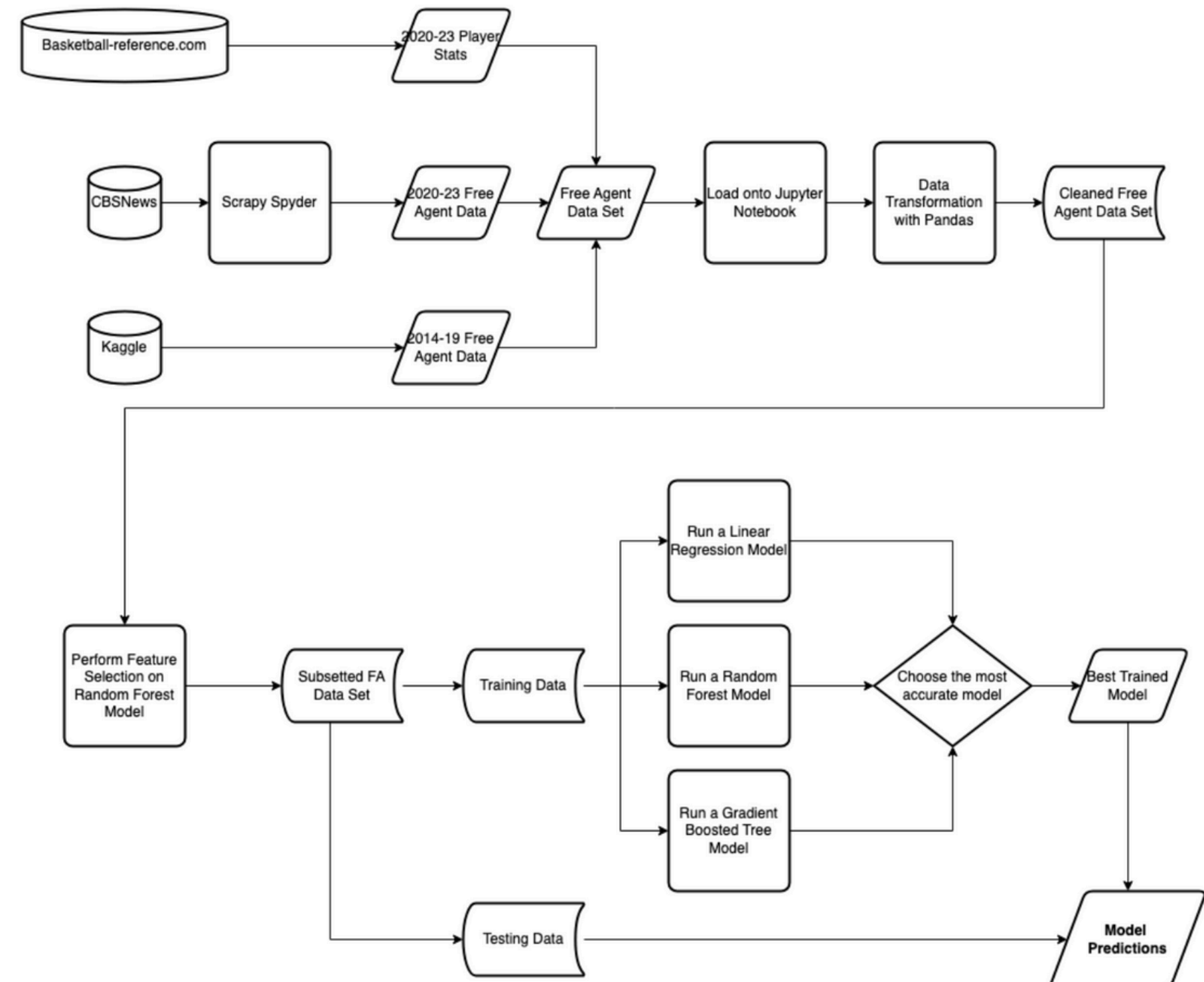


PROCESS

Objective

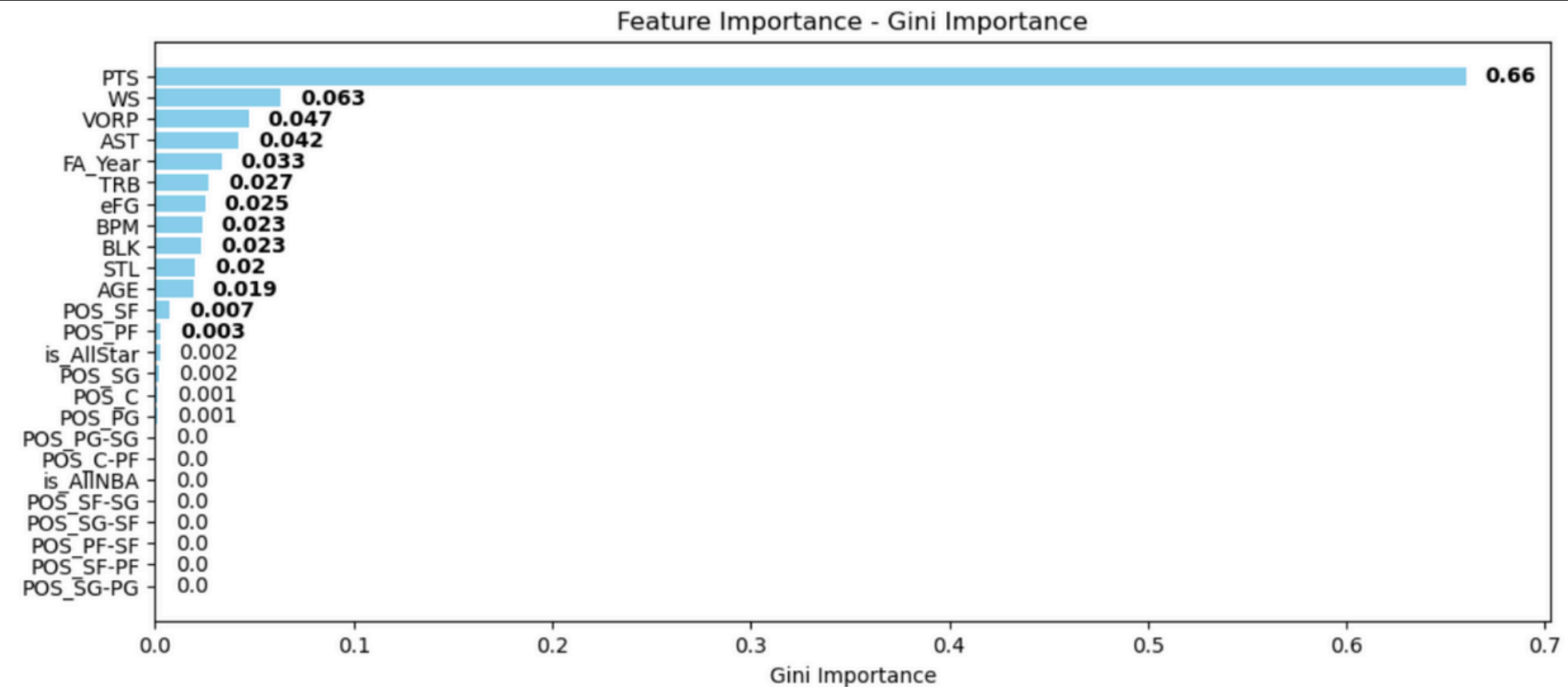
Our objective is to build an NBA dashboard that GMs can use to evaluate offseason decisions and identify valuable assets for their teams

- Scraped data from multiple sites
- Cleaned data through packages in Python and R
- Used past data to train model through statistical modelling methods
- Ran model on current free agents
- Visualized results and provided suggestions



MODEL RESULTS

Valuing variable importance for model building



	Linear Regression	Random Forest	Gradient Boosted
R^2	0.580	0.632	0.608
RMSE	5.50M	5.14 M	5.32 M

Assessing model performance

MODEL SUGGESTIONS

Example Dashboard – Houston Rockets

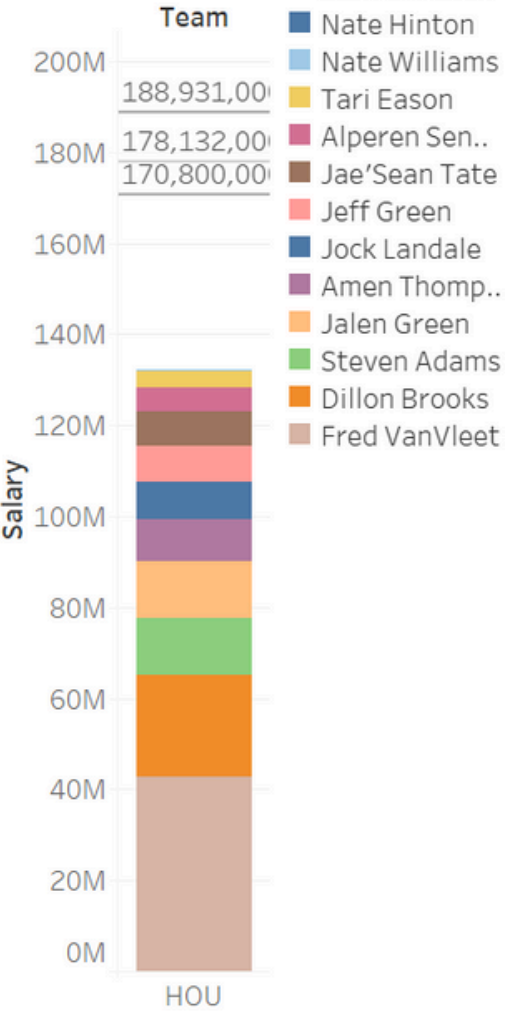
Rockets Pivot Table

		Pos				
	C	PF	PG	SF	SG	
	3	2	1	3	1	
	26.000	16.300	17.400	26.300	19.600	
	12.400	9.300	3.800	13.000	5.200	
	6.200	2.100	8.100	5.300	3.500	
	9.200	3.900	8.400	8.800	3.100	
	3.900	0.400	3.600	0.300	1.000	
	26,024,654	11,695,160	42,846,615	39,070,458	12,483,048	

Free Agents

Player	POS	
LeBron James	PF	25.70
		36,345,484
DeMar DeRozan	SF	24.00
		36,589,362
Pascal Siakam	PF	21.70
		31,548,390
Miles Bridges	SF	21.00
		23,641,882
Tobias Harris	PF	17.20
		21,153,867
Kelly Oubre Jr.	SF	15.40
		16,680,804

Rockets Payroll



Single Season Contract Values

Weakest position – PF

Possible Signings

player	format(expected_cont, 'N')	pts	trb	ast	stl	blk	ws
Dario Saric	7,430,257	8	4.4	2.3	0.5	0.2	2.8

player	format(expected_cont, 'N')	pts	trb	ast	stl	blk	ws
Tobias Harris	21,153,867	17.2	6.5	3.1	1	0.7	5.9

Recommended Signing

player	format(expected_cont, 'N')	pts	trb	ast	stl	blk	ws
Precious Achiuwa	7,616,809	7.6	6.6	1.3	0.6	0.9	3.4

With a low expected contract, Achiuwa provides depth and consistency at the Power Forward position. His youth pairs him well alongside the young talent in Houston.



MODEL CONCLUSIONS

The model does a good job of estimating a player's worth, and combining that with the dashboard's of teams allows NBA GMs to quickly identify a team's weaknesses and strengths and find a suitable player to sign to fill that position. As with all analytical models, the model does not account for a player's fit to a team's system. We believe the model still provides value to a GM because it allows them to see possible and feasible options to choose from. In combination with effective scouting and analysis of free agents, this dashboard allows GMs to efficiently identify valuable assets.

DATA SOURCES

- https://www.basketball-reference.com/leagues/NBA_2024_per_game.html
 - Regular and advanced stats
- <https://www.nba.com/news/2024-nba-free-agents>
 - List of free agents this offseason
- https://www.basketball-reference.com/teams/ATL/2024_depth.html
 - Depth chart, used for all 30 teams
- <https://www.kaggle.com/datasets/jarosawjaworski/current-nba-players-contracts-history/code>
 - 2014-19 Free Agent Dataset
- <https://www.cbssports.com/nba/news/2020-nba-free-agency-tracker-anthony-davis-re-signs-with-lakers-bogdan-bogdanovic-joins-hawks/>
 - 2020 Free Agent Dataset
- <https://www.cbssports.com/nba/news/nba-free-agency-tracker-2021-lakers-reload-with-carmelo-anthony-dwight-howard-other-vets-bulls-make-moves/>
 - 2021 Free Agent Dataset
- <https://www.cbssports.com/nba/news/2022-nba-free-agency-tracker-james-harden-76ers-agree-to-two-year-deal-raptors-sign-juancho-hernangomez/>
 - 2022 Free Agent Dataset
- <https://www.cbssports.com/nba/news/2023-nba-free-agency-tracker-latest-deals-signings-with-notable-names-still-available/>
 - 2023 Free Agent Dataset
- <https://hoopshype.com/salaries/players/>
 - 2024-29 Player Salaries